



US 20250281237A1

(19) **United States**

(12) **Patent Application Publication**
Brannan et al.

(10) **Pub. No.: US 2025/0281237 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **CHOKED DIELECTRIC LOADED TIP
DIPOLE MICROWAVE ANTENNA**

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(21) Appl. No.: **19/214,342**

(22) Filed: **May 21, 2025**

Publication Classification

(51) **Int. Cl.**

A61B 18/18 (2006.01)

A61B 18/00 (2006.01)

H01Q 9/16 (2006.01)

(52) **U.S. Cl.**

CPC **A61B 18/1815** (2013.01); **A61B 18/18**
(2013.01); **H01Q 9/16** (2013.01); **A61B**
2018/00023 (2013.01); **A61B 2018/00071**
(2013.01); **A61B 2018/00166** (2013.01); **A61B**
2018/1838 (2013.01)

Related U.S. Application Data

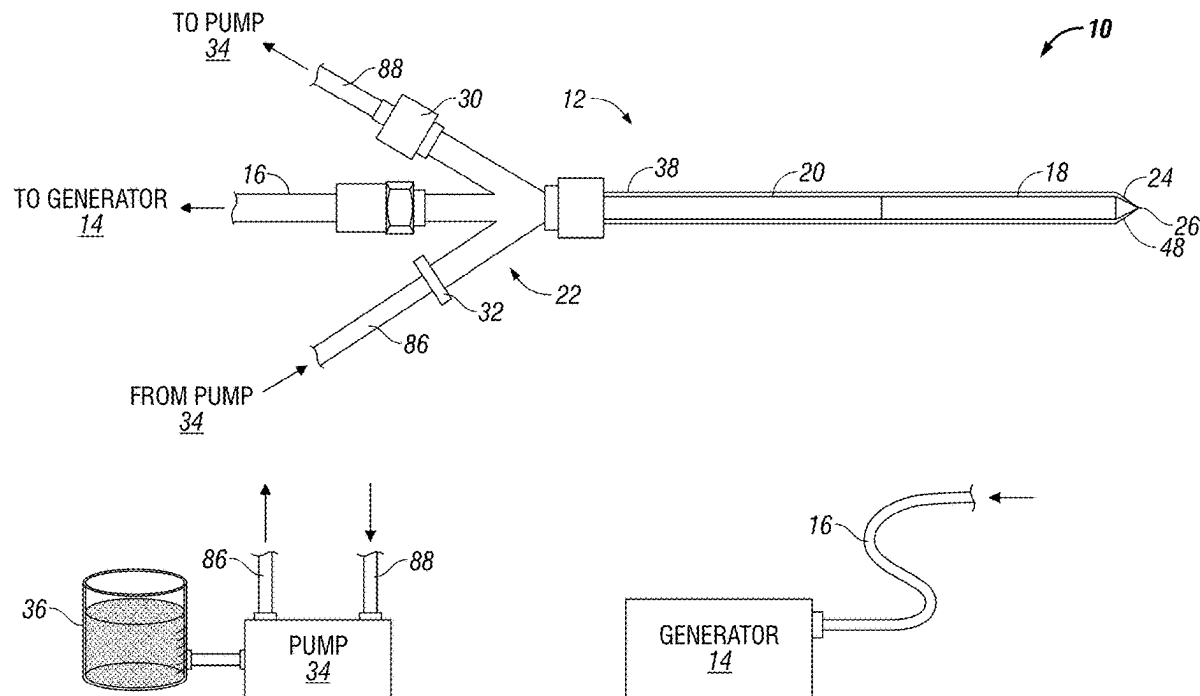
(63) Continuation of application No. 16/987,476, filed on Aug. 7, 2020, now Pat. No. 12,318,133, which is a continuation of application No. 16/113,540, filed on Aug. 27, 2018, now Pat. No. 10,743,934, which is a continuation of application No. 14/599,619, filed on Jan. 19, 2015, now Pat. No. 10,058,384, which is a continuation of application No. 13/444,496, filed on Apr. 11, 2012, now Pat. No. 9,861,439, which is a continuation of application No. 12/350,292, filed on Jan. 8, 2009, now Pat. No. 8,945,111.

(60) Provisional application No. 61/023,031, filed on Jan. 23, 2008.

(57)

ABSTRACT

A microwave antenna assembly is disclosed. The antenna assembly includes a feedline having an inner conductor, an outer conductor and an inner insulator disposed therebetween. A radiating portion is also included having an unbalanced dipole antenna including a proximal portion and a distal portion that are of different lengths. The proximal portion includes at least a portion of the inner conductor and the inner insulator and the distal portion includes a conductive member.



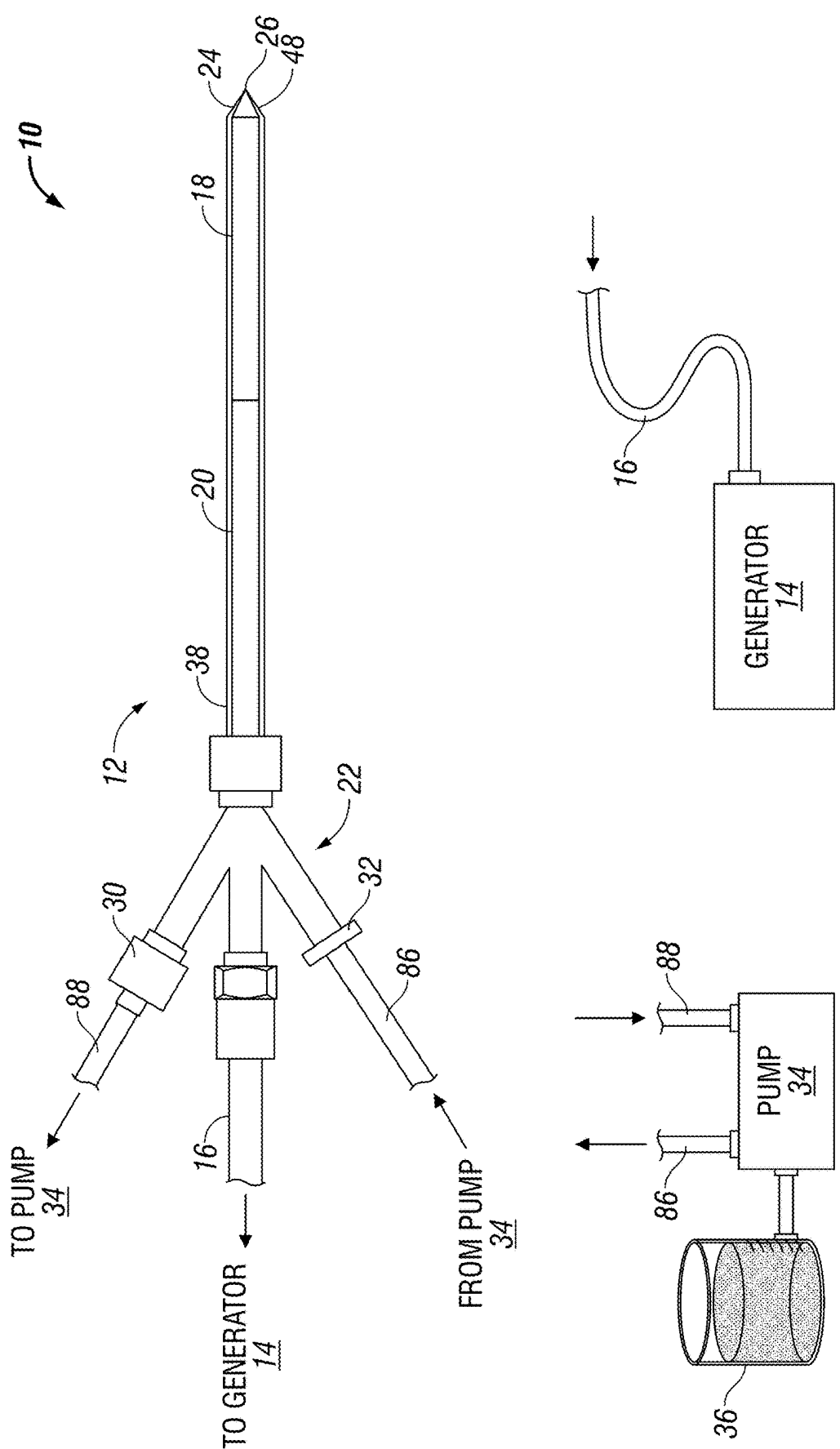


FIG. 1

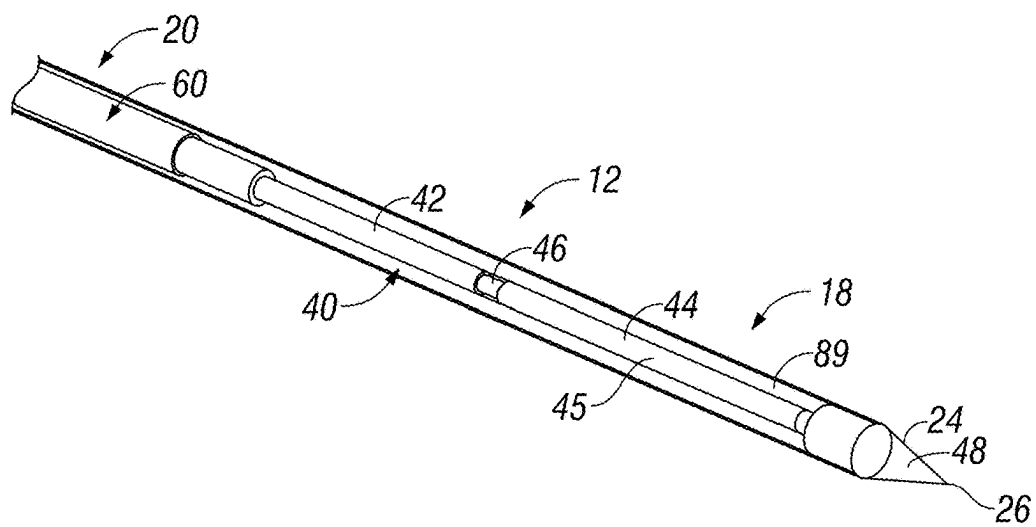


FIG. 2

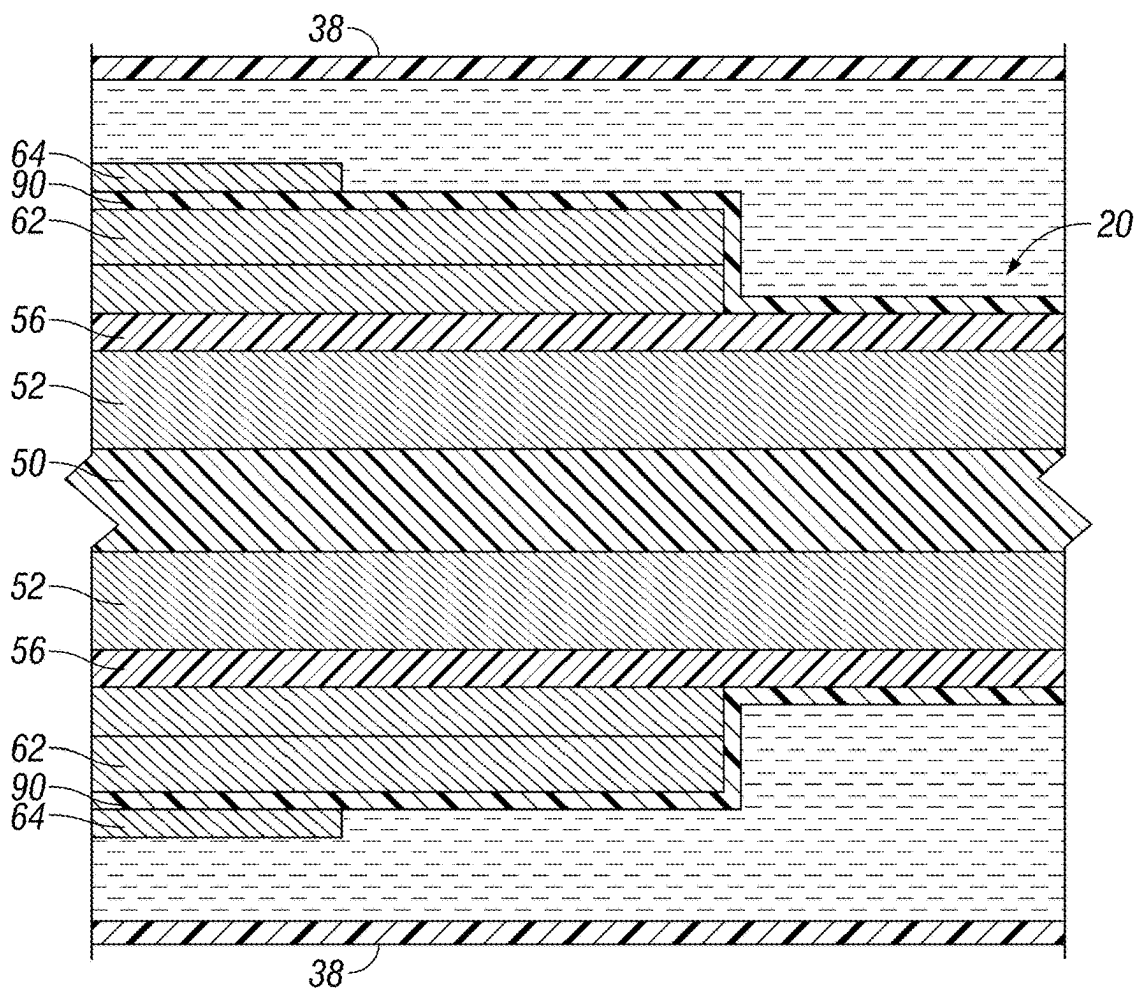


FIG. 3

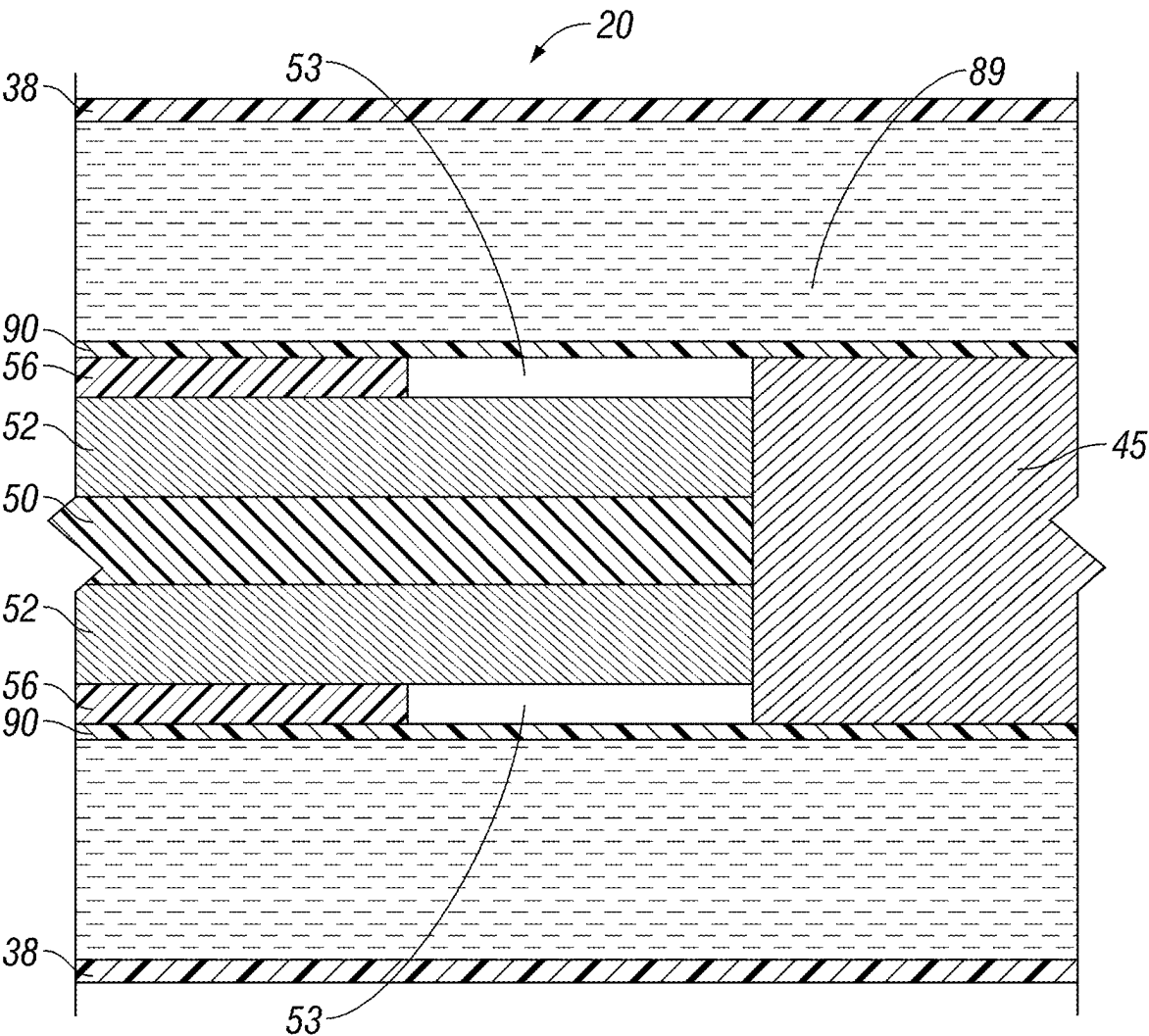


FIG. 4

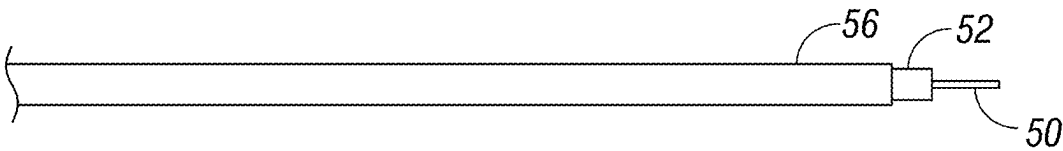


FIG. 5

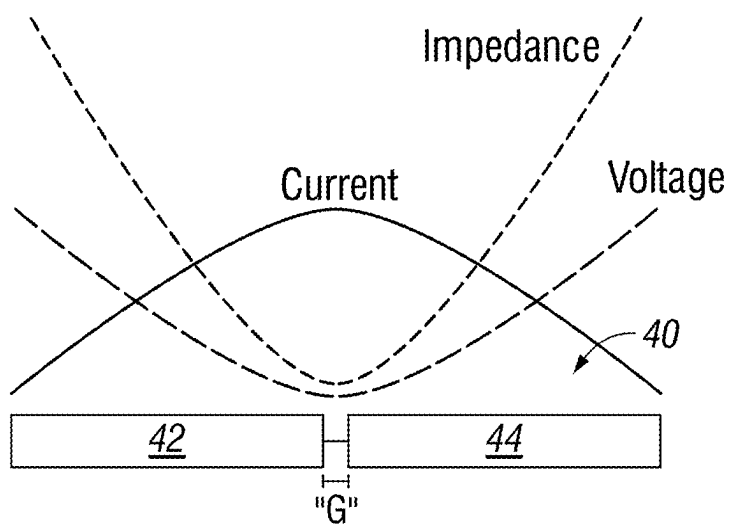


FIG. 6

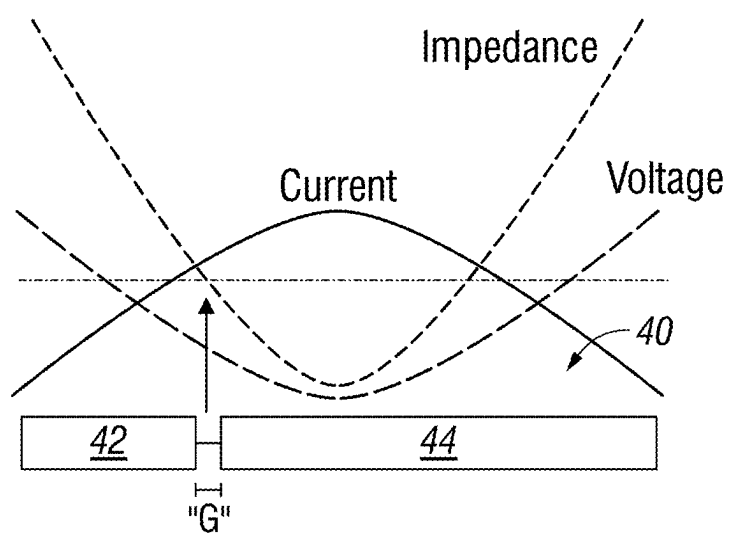


FIG. 7

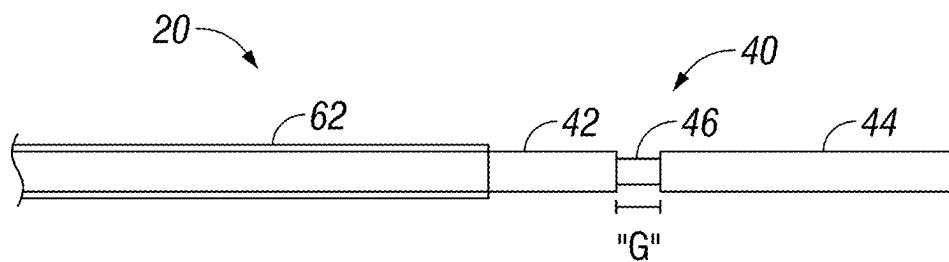
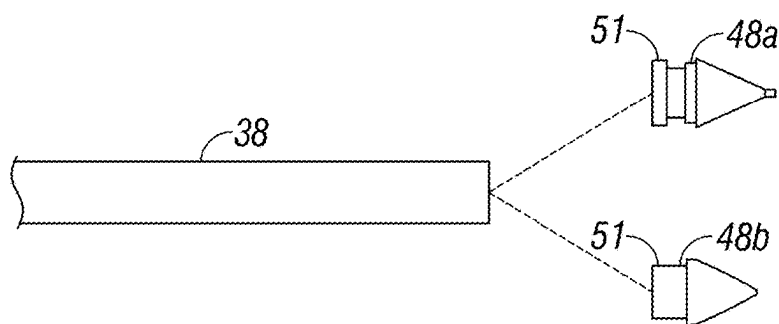
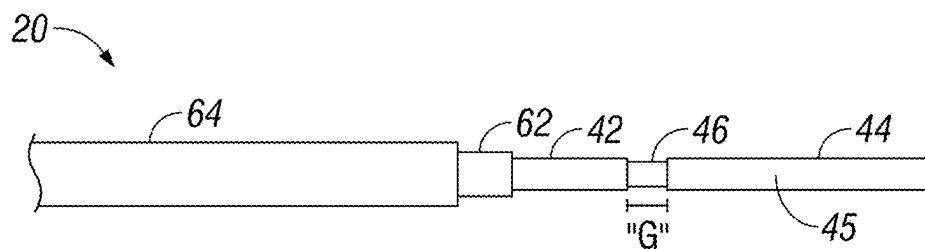
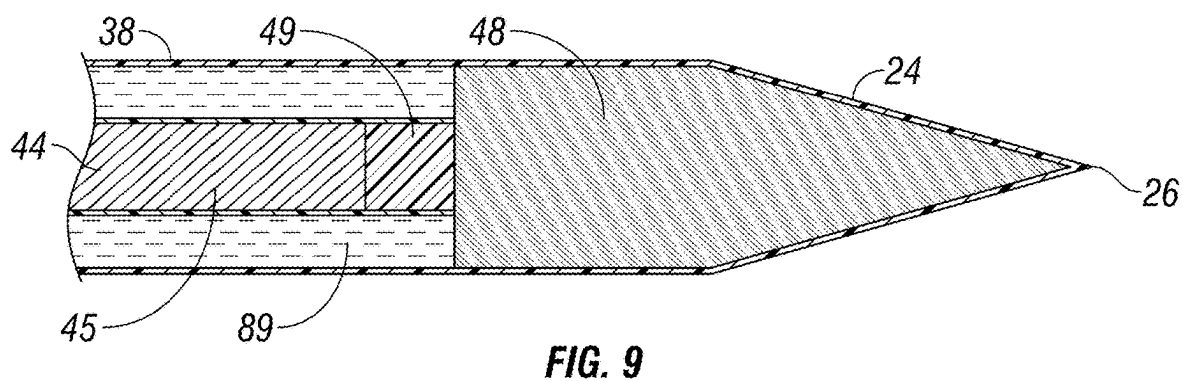


FIG. 8



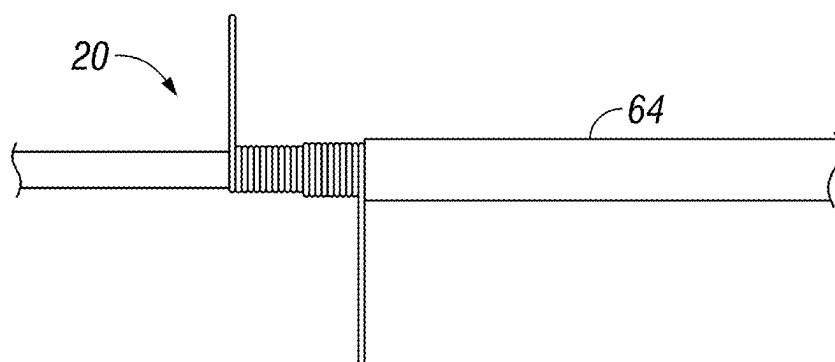


FIG. 12

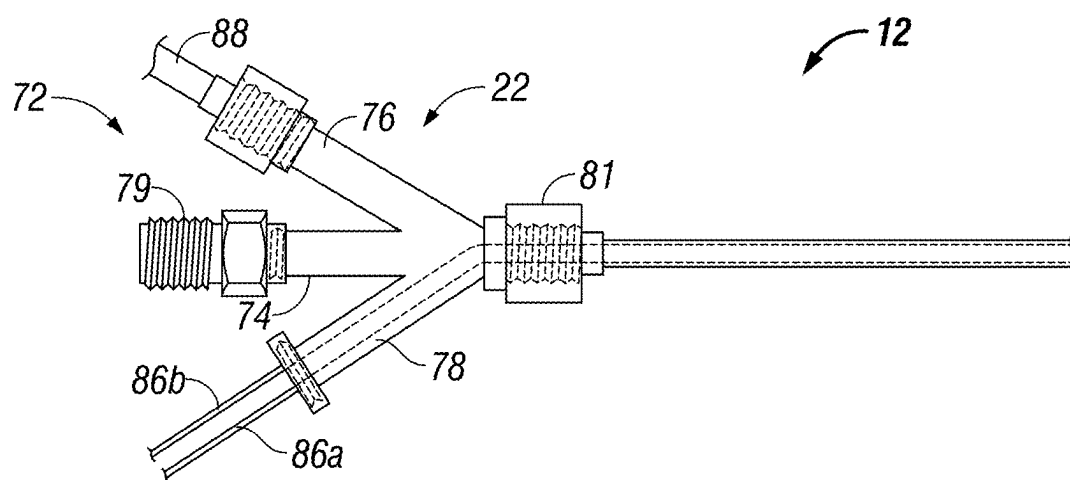


FIG. 13

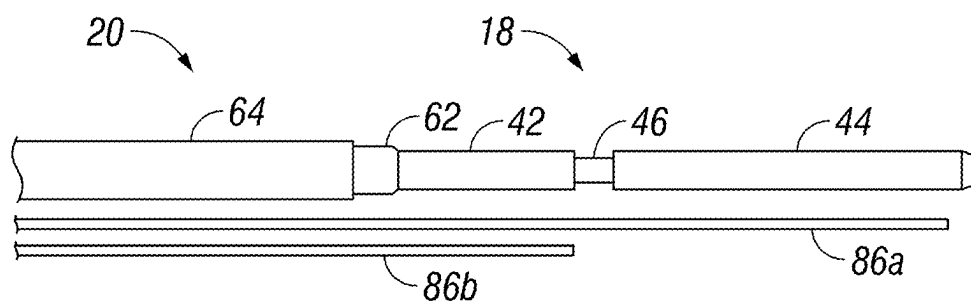


FIG. 14

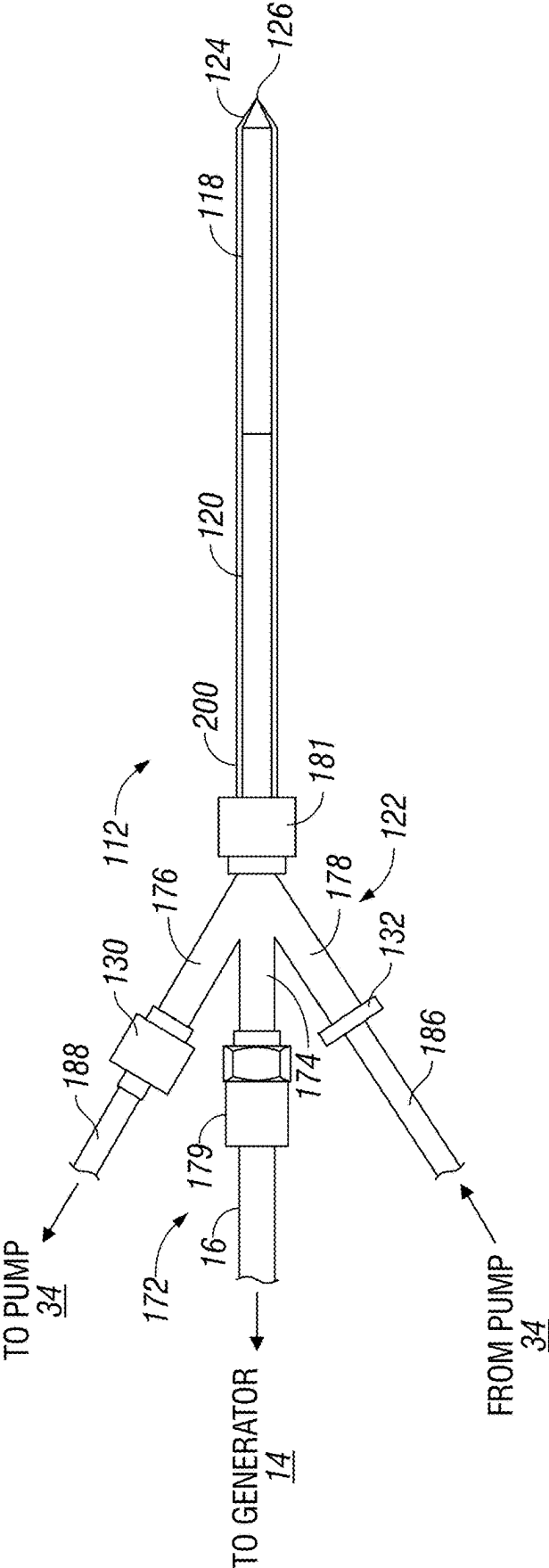


FIG. 15

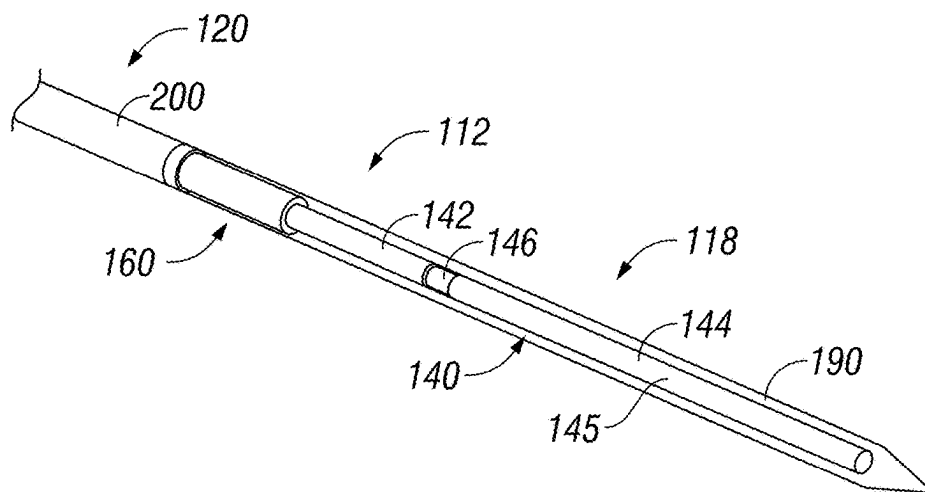


FIG. 16

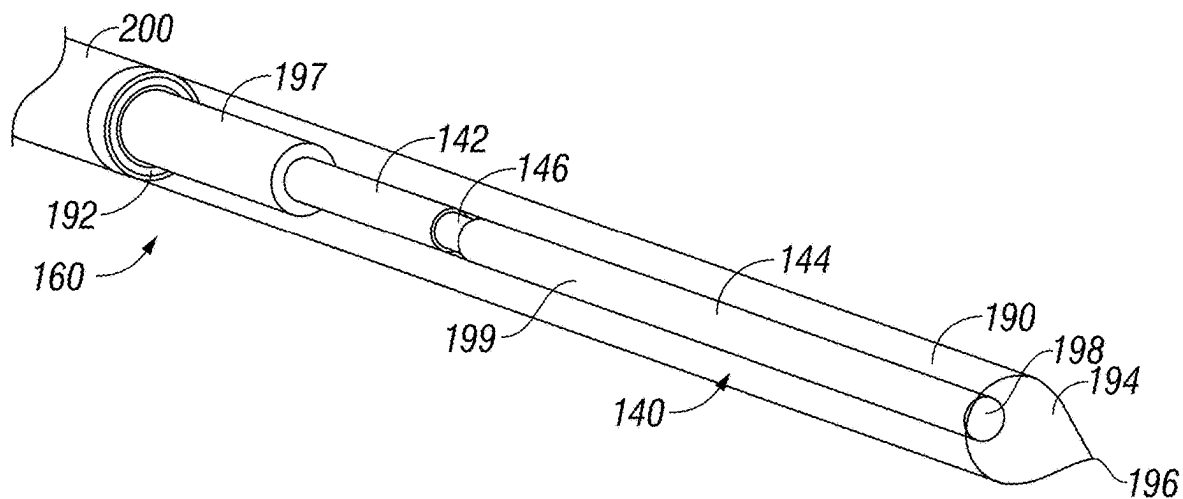


FIG. 17

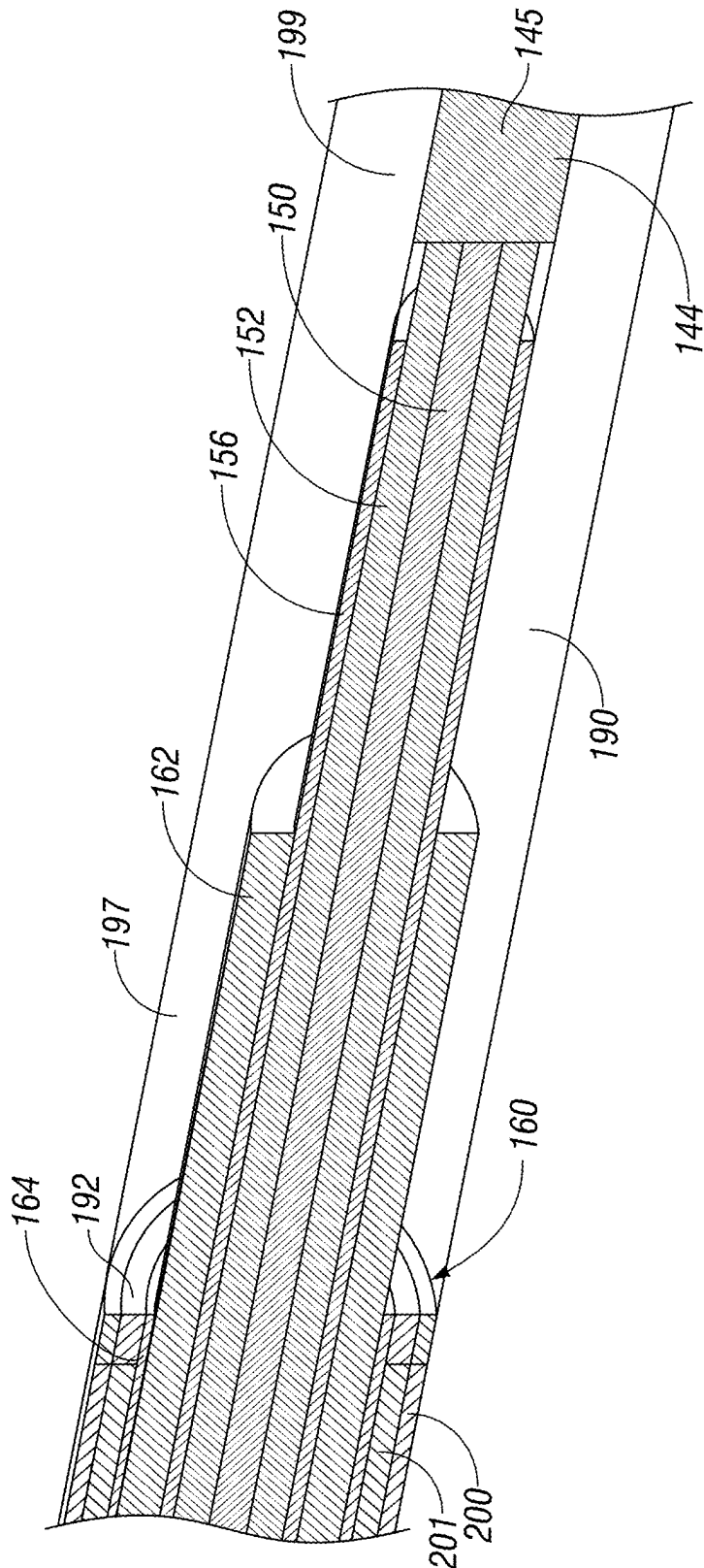


FIG. 18

CHOKED DIELECTRIC LOADED TIP DIPOLE MICROWAVE ANTENNA

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 16/987,476, filed on Aug. 7, 2020, which is a continuation of U.S. application Ser. No. 16/113,540, filed on Aug. 27, 2018, now U.S. Pat. No. 10,743,934, which is a continuation of U.S. application Ser. No. 14/599,619, filed on Jan. 19, 2015, now U.S. Pat. No. 10,058,384, which is a continuation of U.S. application Ser. No. 13/444,496, filed on Apr. 11, 2012, now U.S. Pat. No. 9,861,439, which is a continuation of U.S. application Ser. No. 12/350,292 filed on Jan. 8, 2009, now U.S. Pat. No. 8,945,111, which claims the benefit of the priority date of provisional U.S. Patent Application No. 61/023,031, filed on Jan. 23, 2008.

INTRODUCTION

[0002] The present disclosure relates generally to microwave applicators used in tissue ablation procedures. More particularly, the present disclosure is directed to a microwave applicator having either a liquid or solid loaded tip dipole antenna.

BACKGROUND

[0003] Treatment of certain diseases requires destruction of malignant tissue growths (e.g., tumors). It is known that tumor cells denature at elevated temperatures that are slightly lower than temperatures injurious to surrounding healthy cells. Therefore, known treatment methods, such as hyperthermia therapy, heat tumor cells to temperatures above 41° C., while maintaining adjacent healthy cells at lower temperatures to avoid irreversible cell damage. Such methods involve applying electromagnetic radiation to heat tissue and include ablation and coagulation of tissue. In particular, microwave energy is used to coagulate and/or ablate tissue to denature or kill the cancerous cells.

[0004] Microwave energy is applied via microwave ablation antennas that penetrate tissue to reach tumors. There are several types of microwave antennas, such as monopole and dipole. In monopole and dipole antennas, microwave energy radiates perpendicularly from the axis of the conductor. A monopole antenna includes a single, elongated microwave conductor. Dipole antennas may have a coaxial construction including an inner conductor and an outer conductor separated by a dielectric portion. More specifically, dipole microwave antennas may have a long, thin inner conductor that extends along a longitudinal axis of the antenna and is surrounded by an outer conductor. In certain variations, a portion or portions of the outer conductor may be selectively removed to provide for more effective outward radiation of energy. This type of microwave antenna construction is typically referred to as a “leaky waveguide” or “leaky coaxial” antenna.

[0005] Conventional microwave antennas have a narrow operational bandwidth, a wavelength range at which optimal operational efficiency is achieved, and hence, are incapable of maintaining a predetermined impedance match between the microwave delivery system (e.g., generator, cable, etc.) and the tissue surrounding the microwave antenna. More specifically, as microwave energy is applied to tissue, the

dielectric constant of the tissue immediately surrounding the microwave antenna decreases as the tissue is cooked. The drop causes the wavelength of the microwave energy being applied to tissue to increase beyond the bandwidth of the antenna. As a result, there is a mismatch between the bandwidth of conventional microwave antenna and the microwave energy being applied. Thus, narrow band microwave antennas may detune hindering effective energy delivery and dispersion.

SUMMARY

[0006] According to one aspect of the present disclosure a microwave antenna assembly is disclosed. The antenna assembly includes an unbalanced dipole antenna, a shorted choke having a dielectric layer extending past the conductor layer and connection hub coupled to a coolant system for circulating a dielectric coolant fluid through the antenna assembly.

[0007] According to another aspect of the present disclosure a microwave antenna assembly is disclosed. The antenna assembly includes a feedline having an inner conductor, an outer conductor and an inner insulator disposed therebetween. A radiating portion is included which has an unbalanced dipole antenna having a proximal portion and a distal portion that is longer than the proximal portion. The proximal portion includes at least a portion of the inner conductor and the inner insulator and the distal portion includes a conductive member. The antenna assembly also includes a choke disposed around at least a portion of the feedline. The choke includes an inner dielectric layer and an outer conductive layer, wherein the outer conductive layer is shorted to the outer conductor of the feedline and the inner dielectric layer extends past the outer conductive layer. The assembly further includes a sheath disposed over the feedline and the radiating portion, the sheath defines a chamber around the feedline and the radiating portion, the chamber being adapted to circulate dielectric coolant fluid therethrough.

[0008] According to a further aspect of the present disclosure a microwave antenna assembly is disclosed. The antenna assembly includes a feedline having an inner conductor, an outer conductor and an inner insulator disposed therebetween and a radiating portion including an unbalanced dipole antenna having a proximal portion and a distal portion that are of different lengths. The proximal portion includes at least a portion of the inner conductor and the inner insulator and the distal portion includes a conductive member. The antenna assembly also includes a choke disposed around at least a portion of the feedline. The choke includes an inner dielectric layer and an outer conductive layer, wherein the outer conductive layer is shorted to the outer conductor of the feedline and the inner dielectric layer extends past the outer conductive layer. The antenna assembly further includes a coolant jacket disposed over the feedline defining a proximal chamber around the feedline, the chamber being adapted to circulate dielectric coolant fluid therethrough and a solid dielectric loading having central cavity defined therein adapted to fit about the radiating portion, the solid dielectric loading extending from the coolant jacket.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The above and other aspects, features, and advantages of the present disclosure will become more apparent in

light of the following detailed description when taken in conjunction with the accompanying drawings in which:

[0010] FIG. 1 is a schematic diagram of a microwave ablation system according to an embodiment of the present disclosure;

[0011] FIG. 2 is a perspective cross-sectional view of a microwave antenna assembly according to the present disclosure;

[0012] FIG. 3 is an enlarged cross-sectional of a portion of the microwave antenna assembly of FIG. 2;

[0013] FIG. 4 is an enlarged cross-sectional of a portion of the microwave antenna assembly of FIG. 2;

[0014] FIG. 5 is a side view of a distal portion of a feedline of the microwave antenna assembly of FIG. 2;

[0015] FIG. 6 is a schematic illustration of a balanced dipole antenna according to an embodiment of the present disclosure;

[0016] FIG. 7 is a schematic illustration of an unbalanced dipole antenna according to an embodiment of the present disclosure;

[0017] FIG. 8 is a side view of the unbalanced dipole antenna of the microwave antenna assembly of FIG. 2;

[0018] FIG. 9 is an enlarged cross-sectional of a distal end of the microwave antenna assembly of FIG. 2;

[0019] FIG. 10 is a side view of a radiating portion of the microwave antenna assembly of FIG. 2;

[0020] FIG. 11 is a side view of a tip and a sheath of the microwave antenna assembly of FIG. 2;

[0021] FIG. 12 is a side view of its proximal end of the feedline of the microwave antenna assembly of FIG. 2;

[0022] FIG. 13 is a cross-sectional view of the connection hub and a proximal end of the microwave antenna assembly of FIG. 2;

[0023] FIG. 14 is a schematic view of inflow tubes of the microwave antenna assembly of FIG. 2;

[0024] FIG. 15 is a side view of a microwave antenna assembly according to one embodiment of the present disclosure;

[0025] FIGS. 16 and 17 are perspective cross-sectional views of the microwave antenna of FIG. 15; and

[0026] FIG. 18 is a cross-sectional enlarged perspective view of the microwave antenna of FIG. 15.

DETAILED DESCRIPTION

[0027] Particular embodiments of the present disclosure will be described herein below with reference to the accompanying drawings. In the following description, well-known functions or constructions are not described in detail to avoid obscuring the present disclosure in unnecessary detail.

[0028] FIG. 1 shows a microwave ablation system 10 that includes a microwave antenna assembly 12 coupled to a microwave generator 14 via a flexible coaxial cable 16. The generator 14 is configured to provide microwave energy at an operational frequency from about 500 MHz to about 5000 MHz.

[0029] The antenna assembly 12 is generally comprised of radiating portion 18, which may be connected by feedline 20 (or shaft) to the cable 16. More specifically, the antenna assembly 12 is coupled to the cable 16 through a connection hub 22. The connection hub 22 also includes an outlet fluid port 30 and an inlet fluid port 32 that are connected in fluid communication with a sheath 38. The sheath 38 encloses the radiating portion 18 and the feedline 20 allowing for coolant fluid from the ports 30 and 32 to be supplied and circulated

around the antenna assembly 12. The ports 30 and 32 are also coupled to a supply pump 34 that is, in turn, coupled to a supply tank 36. The supply tank 36 stores the coolant fluid and maintains the fluid at a predetermined temperature. In one embodiment, the supply tank 36 may include a coolant unit which cools the returning liquid from the antenna assembly 12. In another embodiment, the coolant fluid may be a gas and/or a mixture of fluid and gas.

[0030] Assembly 12 also includes a tip 48 having a tapered end 24 that terminates, in one embodiment, at a pointed end 26 to allow for insertion into tissue with minimal resistance at a distal end of the radiating portion 18. In those cases where the radiating portion 18 is inserted into a pre-existing opening, tip 48 may be rounded or flat.

[0031] FIG. 2 illustrates the radiating portion 18 of the antenna assembly 12 having an unbalanced dipole antenna 40. The dipole antenna 40 is coupled to the feedline 20 that electrically connects antenna assembly 12 to the generator 14. As shown in FIG. 3-4, the feedline 20 includes an inner conductor 50 (e.g., wire) surrounded by an inner insulator 52, which is then surrounded by an outer conductor 56 (e.g., cylindrical conducting sheath). The inner and outer conductors may be constructed of copper, gold, stainless steel or other conductive metals with similar conductivity values. The metals may be plated with other materials, e.g., other conductive materials, to improve their properties, e.g., to improve conductivity or decrease energy loss, etc. In one embodiment, the feedline 20 may be formed from a coaxial semi-rigid or flexible cable having a wire with a 0.047" outer diameter rated for 50 Ohms.

[0032] The dipole antenna 40 includes a proximal portion 42 and a distal portion 44 interconnected by a dielectric spacer at a feed point 46. The distal portion 44 and the proximal portion 42 are of different, unequal lengths so that the dipole antenna 40 is unbalanced. In one embodiment, as shown in FIG. 7, the distal portion 44 may be longer than the proximal portion 42. The proximal portion 42 is formed from the inner conductor 50 and the inner insulator 52 which are extended outside the outer conductor 56, as shown best in FIG. 4. In one embodiment, in which the feedline 20 is formed from a coaxial cable, the outer conductor 56 and the inner insulator 52 may be sliced off to reveal the inner conductor 50, as shown in FIG. 5.

[0033] The distal portion 44 includes a conductive member 45 that may be formed from any type of conductive material, such as metals (e.g., copper, stainless steel, tin, and various alloys thereof). The distal portion 44 may have a solid structure and may be formed from solid wire (e.g., 10 AWG). In another embodiment, the distal portion 44 may be formed from a hollow sleeve of an outer conductor of coaxial cable or another cylindrical conductor. The cylindrical conductor may then be filled with solder to convert the cylinder into a solid shaft. More specifically, the solder may be heated to a temperature sufficient to liquefy the solder within the cylindrical conductor (e.g., 500° F.) thereby creating a solid shaft.

[0034] In another embodiment, the conductive member 45 may also be formed from solid wire or a cylindrical conductor filled with solder. The conductive member 45 is thereafter coupled to the inner conductor 50, as shown in FIG. 4. This may be accomplished by soldering the conductive member 45 to the distal end of the inner conductor 50, such as by melting the solder of the conductive member 45 and inserting the inner conductor 50 therein.

[0035] In some embodiments, the unbalanced dipole antenna **40** provides for better impedance matching during ablation. Variation in tissue properties during ablation complicates real part impedance matching of microwave ablation antennas. Over the course of an ablation, a given position on the dipole varies in real impedance due to the resulting dynamic current and voltage relationship. FIG. **6** shows the difficulty in matching real part impedance using a half-wave dipole antenna which includes two portions of equal lengths, at the center of the dipole the voltage is minimized and the current is maximized. However, the real part impedance is minimized and is maximized at the ends of the proximal and distal portions **42** and **44**. In contrast, the unbalanced dipole antenna **40** of the present disclosure minimizes the integration over ablation time of the difference between the feed point real part impedance and the impedance of the cable **16**. As illustrated in FIG. **7**, the unbalanced half-wave dipole provides a better match of initial impedance to real part impedance by placing the gap between the proximal and distal portions **42** and **44** away from the center of the dipole antenna **40**. In one embodiment, the length of the distal portion **40** is about 40 mm to minimize return loss of the assembly **12**.

[0036] FIG. **8** illustrates the distal portion **44** attached to the proximal portion **42**. The distal portion **44** may be soldered to the inner conductor **50** of the proximal portion **42** to establish electromechanical contact therebetween. In one embodiment, where the distal portion **44** is formed from a hollow cylindrical conductor filled with a solder material, the distal portion **44** may be attached to the proximal portion **42** by liquefying the solder of the distal portion **44** and inserting the distal end of the inner conductor **50** therein. A portion of the distal end of the inner conductor **50** is inserted into the distal portion **44** such that a dipole feed gap “G” remains between the proximal and distal portions **42** and **44** at the feed point **46**. The gap “G” may be from about 1 mm to about 3 mm. The dipole feed gap of the antenna is the first structure the coaxial field mode encounters upon transfer to free space. The gap therefore plays an important role in the return loss, or system-to-antenna impedance match. In one embodiment, the gap “G” is thereafter filled with a dielectric material to form the dielectric spacer at the feed point **46**. In another embodiment, the inner insulator **52** is extended into the feed point **46**. The dielectric material may be polytetrafluoroethylene (PTFE), such as Teflon® sold by DuPont of Wilmington, DE. In another embodiment, as shown in FIG. **4**, the gap “G” may be coated via a dielectric seal coating as discussed in more detail below.

[0037] As shown in FIGS. **2** and **9**, the distal portion **44** is coupled to the tip **48**, which may be formed from a variety of heat-resistant materials suitable for penetrating tissue, such as metals (e.g., stainless steel) and various thermoplastic materials, such as polyetherimide, polyamide thermoplastic resins, an example of which is Ultem® sold by General Electric Co. of Fairfield, CT. The tip **48** may be machined from various stock rods to obtain a desired shape. The tip **48** may be attached to the distal portion **44** using various adhesives, such as epoxy seal **49**. If the tip **48** is metal, the tip **48** may be soldered to the distal portion **44**.

[0038] FIG. **11** illustrates various shapes and forms of the tip **48**, namely a stainless steel tip **48a** and a dielectric tip **48b**. Both tips **48a** and **48b** includes an insertion base **51** having an external diameter that is smaller than diameter of the tips **48a** and **48b** allowing for easier insertion into the

sheath **38**. This configuration also provides for a better seal between the tip **48** and the sheath **38** as discussed in more detail below.

[0039] With reference to FIGS. **2** and **3**, the antenna assembly **12** also includes a choke **60**. The choke **60** is disposed around the feedline **20** and includes an inner dielectric layer **62** and an outer conductive layer **64**. In one embodiment, the choke **60** is a proximally positioned quarter-wave length shorted choke. The choke **60** is implemented as a quarter-wave length shorted by using the outer conductive layer **64** around the outer conductor **56** of the feedline **20** separated by the dielectric layer. The choke **60** is shorted to the outer conductor **56** of the feedline **20** at the proximal end of the choke **60** by soldering or other means. In one embodiment, the dielectric layer **62** is formed from a fluoropolymer, such as tetrafluorethylene, perfluoropropylene, and the like, and has a thickness of 0.005 inches. The outer conductive layer **64** may be formed from a so-called “perfect conductor” material, such as a highly conductive metal (e.g., copper).

[0040] In embodiments, the choke **60** may be a quarter-wavelength shorted choke, a half-wavelength open choke, and inverted quarter-wavelength shorted choke or a gap cancellation choke. The choke **60** confines the microwave energy from the generator **14** to the radiating portion **18** of the assembly **12**, thereby limiting the microwave energy deposition zone length along the feedline **20**. The choke **60** provides high impedance to microwave energy conducted down the outside of the feedline **20**, thereby limiting energy deposition to the end of the antenna.

[0041] A shorted quarter-wave choke placed at the high impedance point of the proximal portion **42** on the antenna assembly **12** confines antenna currents to the radiating section **18** of the assembly **12**, reducing the length and maximizing the cross sectional diameter of ablations due to nearly spherical power dissipation zones.

[0042] The dielectric of dielectric layer **62** extends past the choke conductor layer **64** toward the distal end of the assembly **12**, as shown in FIG. **10**. In one embodiment, the dielectric layer **62** may extend past the choke conductor layer **64** by about 6 mm. This extended dielectric improves the performance of the choke **60** by placing a capacitance between the proximal portion **42** of the dipole and the outer surface of the choke conductor layer **64** thereby blocking currents from jumping onto the choke conductor layer **64**. The capacitance formed by the dielectric is a high impedance barrier to microwave currents which would otherwise jump from the proximal portion **42** to the outer surface of the choke **60** near the entrance thereof, avoiding the choke structure completely. Instead, these currents are directed into the quarter-wave choke **60** by the capacitance, improving its effectiveness.

[0043] As discussed above, the wavelength increase due to tissue desiccation causes the high impedance point on the proximal portion **42** to move proximally along the assembly **12**. An effective choke must present high impedance at this variable point. The extended dielectric effectively acts as a variable position choke, covering the range over which this point shifts, maintaining choke effectiveness as long as the high impedance point of the proximal portion **42** stays within the extended dielectric boundaries. The dielectric layer **62** may be extended to any length between the choke conductive layer **64** and the feed point **46**.

[0044] In one embodiment, the dielectric layer 62 may be formed by applying a dielectric shrink material, such as 5/64" thick PTFE shrink wrap to the outer conductor 56. Once the shrink wrap material is placed around the outer conductor 56, the material is heated so that the material melts and sets about the outer conductor 56. The heating may be accomplished by hot air blowers, which can provide a hot air stream of about 750° F. Multiple layers of the PTFE shrink wrap may be applied and consecutively heated to form the dielectric layer 62 of desired thickness. In one embodiment, three or more layers of the PTFE shrink wrap are applied.

[0045] As shown in FIGS. 3 and 10, the conductor layer 64 may be formed by applying one or more layers of a conductive metal foil (e.g., copper) onto the dielectric layer 62. The foil may extend past the proximal end of the dielectric layer 62 as shown in FIG. 12. The foil may be attached to the dielectric layer 62 using various types of adhesives (e.g., ultraviolet light activated glue, epoxy, etc.). In one embodiment, the proximal end of the foil which extends past the dielectric layer 62 may be attached to the feedline 20 by means of a so-called "wire-wrap" technique to provide a good electrical connection to the foil and the feedline 20 as shown in FIG. 12. The wire is wrapped around the copper foil at the point where the foil begins to taper down past the dielectric layer 62. After the wire is wrapped, the wire is soldered to itself all along the length of the wrap to secure the wire and prevent the wire from unwrapping. In another embodiment, other means may be used to secure the foil to the feedline 20, such as a hollow cylinder may be placed around the excess foil necking down past the dielectric layer 62. In a further embodiment, the foil may be substantially the same length as the dielectric layer 62 to obviate the need for securing the proximal end of the foil to the feedline 20.

[0046] The assembly 12 also includes the connection hub 22 of FIG. 1, as shown in more detail in FIG. 13. The connection hub 22 includes a cable connector 79 and fluid ports 30 and 32. The connection hub 22 may include a three-branch luer type connector 72, with a middle finger 74 being used to house the cable connector 70 and the left and right fingers 76 and 78 to house the outlet and inlet fluid ports 30 and 32, respectively. The connection hub 22 also includes a base 81 disposed at a distal end of the middle finger 74.

[0047] The assembly 12 also includes an active coolant system as shown in FIGS. 1, 13 and 14. More specifically, the assembly 12 includes sheath 38 that encloses the feedline 20, the radiating portion 18 from the tip 48 to the base 81. The coolant is supplied by the pump 34 and is circulated in the space between the radiating portion 18, the feedline 20 and the sheath 38. Since the radiating portion 18 and the feedline 20 are in direct contact with the coolant fluid these components of the assembly 12 should be sealed to prevent any fluid seeping therein. This may be accomplished by applying any type of melt-processible polymers using conventional injection molding and screw extrusion techniques. In one embodiment, a sleeve 90 of fluorinated ethylene propylene (FEP) shrink wrap as shown in FIGS. 3 and 4 may be applied to the entire assembly 12, namely the feedline 20 and the radiating portion 18, as shown in FIG. 1. The FEP sleeve 90 is then heated to seal the feedline 20 and radiating portion 18. The resulting FEP seal prevents any coolant fluid from penetrating into the assembly 12. The FEP sleeve 90

may be applied either prior to (FIG. 3) or after applying the outer conductive layer 64. In addition, FEP may also be applied at the point where the inner conductor 50 and the inner insulator 52 are extended past the outer conductor 56, thereby creating a vacuum 53 as shown in FIG. 4.

[0048] The sheath 38 may be any type of rigid tube, such as a catheter manufactured from polyimide and other types of polymers. The sheath 38 may be assembled by initially securing the tip 48 to the distal end of the sheath 38 and then inserting the combined sheath and tip assembly onto the assembly 12. The sheath 38 is also secured to the base 81 of the connection hub 22 and the tip 48 such that the sheath 38 is in fluid communication with the connection hub 22 and defines a chamber 89 between the base 81 and the tip 48.

[0049] The inflow tube 86 may include one or more inflow tubes 86a and 86b. The inflow tubes 86a and 86b may be any type of flexible tube having an external diameter sufficient to fit inside the chamber 89 (FIGS. 4 and 9) between the feedline 20 and the sheath 38. The inflow tubes 86a and 86b are inserted through the outlet fluid port 30. More specifically, the inflow tube 86a is inserted almost to the distal end of the distal portion 44 and the inflow tube 86b is inserted approximately to the feed point 46 as shown in FIG. 14. The inflow tubes 86a and 86b are then secured to the radiating portion 18 (e.g., using epoxy, glue, etc.). The inflow tubes 86a and 86b are positioned in this configuration to provide for optimal coolant flow through the sheath 38. The fluid flow from the inflow tube 86a is ejected into the tip 48 and is reflected in the proximal direction. The fluid flow from the inflow tube 86b provides for the coolant along the radiating portion 18. During operation, the pump 34 supplies fluid to the assembly 12 through the inflow tubes 86a and 86b, thereby circulating the coolant through the entire length of the assembly 12 including the connection hub 22. The fluid is then withdrawn from the middle finger 74 and the left finger 76 through the outlet fluid port 32.

[0050] The above-discussed coolant system provides for circulation of dielectric coolant fluid (e.g., saline, deionized water, etc.) through the entire length of the antenna assembly 12. The dielectric coolant fluid removes the heat generated by the assembly 12. In addition, the dielectric coolant fluid acts as a buffer for the assembly 12 and prevents near field dielectric properties of the assembly 12 from changing due to varying tissue dielectric properties. As microwave energy is applied during ablation, desiccation of the tissue around the radiating portion 18 results in a drop in tissue complex permittivity by a considerable factor (e.g., about 10). The dielectric constant (ϵ_r') drop increases the wavelength of microwave energy in the tissue, which dramatically affects the impedance of un-buffered microwave antenna assemblies, thereby mismatching the antenna assemblies from the system impedance (e.g., impedance of the cable 16 and the generator 14). The increase in wavelength also results in a power dissipation zone which is much longer in length along the assembly 12 than in cross sectional diameter. The decrease in tissue conductivity (ϵ_r'') also affects the real part of the impedance of the assembly 12. The fluid dielectric buffering according to the present disclosure also moderates the increase in wavelength of the delivered energy and drop in conductivity of the near field, thereby reducing the change in impedance of the assembly 12, allowing for more consistent antenna-to-system impedance match and spherical power dissipation zone despite tissue behavior.

[0051] The buffering of wavelength variation also allows for a more effective choking network. The choke must be placed at the low current point, or high impedance point, on the end of the proximal portion 42. With wavelength buffering in the choked wet tip, the half wavelength current pattern on the dipole radiating section is maintained, making the position of the high impedance point less variable and therefore allowing for a more effective choke network. Together, the cable cooling and the dielectric buffering allow for targeted and efficient energy delivery to the tissue to enable nearly spherical ablation zones and fast ablation times. Either saline or deionized water can be used with the assembly 12.

[0052] FIGS. 15-18 illustrate another embodiment of a microwave antenna assembly 112 of having a radiating portion 118 and a feedline 120 which couples the assembly 112 to the cable 16. More specifically, the antenna assembly 112 is coupled to the cable 16 through a connection hub 122 that includes an outlet fluid port 130 and an inlet fluid port 132.

[0053] FIGS. 16 and 17 illustrate the radiating portion 118 of the antenna assembly 112 having an unbalanced dipole antenna 140 in which the sheath 38 is replaced by a metallic conduit (e.g., coolant jacket 200) and a solid dielectric loading 190. The dipole antenna 140 is coupled to the feedline 120, which electrically connects antenna assembly 112 to the generator 14. As shown in FIG. 18, similar to the feedline 20, the feedline 120 includes an inner conductor 150 (e.g., wire) surrounded by an inner 152 insulator which is then surrounded by an outer conductor 156 (e.g., cylindrical conducting sheath).

[0054] The dipole antenna 140 includes a proximal portion 142 and a distal portion 144 interconnected by a dielectric spacer at a feed point 146. The distal portion 144 includes a conductive member 145. The distal portion 144 and the proximal portion 142 are of different, unequal lengths so that the dipole antenna 40 is unbalanced. The proximal portion 142 is formed from the inner conductor 150 and the inner insulator 152 which are extended outside the outer conductor 156. In one embodiment, in which the feedline 120 is formed from a coaxial cable, the outer conductor 156 and the inner insulator 152 may be sliced off to reveal the inner conductor 150 as shown in FIG. 18.

[0055] The distal portion 144 may be formed from any type of conductive material such as metals (e.g., copper, stainless steel, tin, and various alloys thereof). The portion 144 may have a solid structure and may be formed from solid wire (e.g., 10 AWG) or a cylindrical conductor filled with solder similar to the portion 44 of the assembly 12. The proximal portion 144 is thereafter coupled to the inner conductor 150.

[0056] With reference to FIGS. 16-18, the antenna assembly 112 also includes a choke 160. The choke 160 is disposed around the feedline 120 and includes an inner dielectric layer 162 and an outer conductive layer 164. In one embodiment, the choke 160 is a proximally positioned quarter-wave shorted choke that is shorted to the outer conductor 156 of the feedline 120 at the proximal end of the choke 160 by soldering or other means. The dielectric of dielectric layer 162 extends past the choke conductor layer 164 toward the distal end of the assembly 112.

[0057] The assembly 112 also includes the connection hub 122, as shown in FIGS. 15. The connection hub 122 includes a cable connector 179 and the fluid ports 130 and 132. The

connection hub 122 may include a three-branch luer type connector 172, with a middle finger 174 being used to house the cable connector 179 and the left and right fingers 176 and 178 to house the outlet and inlet fluid ports 130 and 132, respectively. The cable connector 179 is coupled to the inner conductor 152 and outer conductor 156 that are extended outside the outer conductor 156 at the proximal end of the feedline 120. The connection hub 122 also includes a base 181 disposed at a distal end of the middle finger 174. In one embodiment, the assembly 112 includes one or more inflow tubes 186 which are fed through the right finger 178.

[0058] The assembly 112 includes a solid dielectric loading 190 disposed over the dipole antenna 140 replacing the liquid dielectric material of assembly 112. The solid dielectric loading 190 extend from the point of termination of the choke conductor layer 164. More specifically, the assembly 112 includes a fluid seal 192 over the distal end of the choke conductor layer 164. In one embodiment, the loading 190 may be attached to the seal 192 via glue and other means.

[0059] The loading 190 may be cylinder-shaped having a central cavity 198 defined therein suitable for insertion over the antenna 140. The loading 190 may also have a tapered end 194 with a pointed tip 196, thereby obviating the need for the tip 48. The loading 190 may also be attached to the distal end of the antenna 140 (e.g., at the distal portion 144 thereof) within the cavity 198. The cavity 198 may have a substantially cylindrical shape suitable to fit over the antenna 140 depending on the cross-sectional shape thereof. In addition, the cavity 198 includes a proximal portion 197 and a distal portion 199 with the proximal portion 197 having a larger inner diameter than the distal portion 199 to accommodate the choke dielectric layer 162. The choke layer 162 may be extended to any length between the choke conductive layer 164 and the feed point 146. To accommodate the extended choke layer 162 the depth of the proximal portion 197 varies accordingly.

[0060] The loading 190 has an outer diameter being substantially equal to the thickness of the feedline 120 and the inner diameter being substantially equal to the diameter of the dipole antenna 140. Since the loading 190 is disposed on the dipole antenna 140 and no coolant fluid is going to be in contact therewith, the antenna 140 may not be coated in dielectric shrink wrap to seal its components.

[0061] In one embodiment, the dielectric material of the loading 90 may have a dielectric constant of from about 2.5 and 150 and may be made from a ceramic material, such as alumina ceramic or a plastic material, such as a polyamide plastic (e.g., VESPEL® available from DuPont of Wilmington, Delaware). The loading 190 acts as a dielectric buffer between the radiating portion 118 and the tissue so that as the electrical properties of the tissue change during ablation the antenna assembly 112 remains halfwave resonant and impedance-matched to the energy delivery system (e.g., the generator 14, the cable 16, etc.) throughout the ablation.

[0062] The antenna assembly 112 also includes a coolant jacket 200 disposed between the base 181 and the seal 192. The coolant jacket 200 maybe formed from stainless steel or other suitable medical grade metals. The coolant jacket 200 defines a proximal chamber 201 between the choke conductor layer 164 and the coolant jacket 200 into which a dielectric coolant fluid is supplied through the connection hub 122. More specifically, one or more inflow tube 186 similar to the tubes 86a and 86b may extend into the chamber 201 to circulate the dielectric coolant fluid through

the coolant jacket **200**. The seal **192** is disposed between the coolant jacket **200** and the choke conductor layer **164** at the distal ends thereof. The seal **192** may be formed from any type of dielectric (e.g., elastomer) and/or conductive material suitable for sealing the chamber **201** from the loading **190**.

[0063] The described embodiments of the present disclosure are intended to be illustrative rather than restrictive, and are not intended to represent every embodiment of the present disclosure. Various modifications and variations can be made without departing from the spirit or scope of the disclosure as set forth in the following claims both literally and in equivalents recognized in law.

1-20. (canceled)

21. A microwave antenna assembly, comprising:

a feedline including an inner conductor, an outer conductor, and an inner insulator disposed between the inner and outer conductors;

a radiating portion including an unbalanced dipole antenna, the unbalanced dipole antenna having a proximal portion, a distal portion, and a gap disposed longitudinally off-center of the unbalanced dipole antenna between the proximal and distal portions, wherein the proximal portion includes at least a portion of the inner conductor and the inner insulator and the distal portion includes a conductive member coupled to a distal end portion of the inner conductor; and

a choke disposed on the feedline, the choke including an inner dielectric layer and an outer conductive layer, wherein the inner dielectric layer extends distally beyond a distalmost end of the outer conductive layer.

22. The microwave antenna assembly according to claim **21**, further comprising a sheath enveloping the radiating portion and defining a fluid chamber for receiving a fluid.

23. The microwave antenna assembly according to claim **22**, further comprising a connection hub including a cable connector coupled to the feedline, an inlet fluid port through which fluid is delivered to the fluid chamber, and an outlet fluid port through which fluid is removed from the fluid chamber.

24. The microwave antenna assembly according to claim **23**, further comprising an inflow tube in fluid communication with the inlet fluid port and at least partially disposed within the fluid chamber.

25. The microwave antenna assembly according to claim **22**, further comprising a tapered distal tip directly coupled to a distal end of the sheath.

26. The microwave antenna assembly according to claim **21**, wherein the outer conductive layer is shorted to the outer conductor of the feedline.

27. The microwave antenna assembly according to claim **21**, wherein a length of the distal portion is different from a length of the proximal portion.

28. The microwave antenna assembly according to claim **21**, wherein a length of the distal portion is greater than a length of the proximal portion.

29. The microwave antenna assembly according to claim **21**, wherein an outermost diameter of the conductive member is greater than an outermost diameter of the inner conductor.

30. A microwave antenna assembly comprising:

a feedline including an inner conductor, an outer conductor, and an inner insulator disposed between the inner and outer conductors;

a radiating portion including an unbalanced dipole antenna having a proximal portion, a distal portion, and a gap disposed between the proximal and distal portions, the gap being disposed at a location along the unbalanced dipole antenna configured to provide an impedance match between the unbalanced dipole antenna and surrounding tissue, wherein the proximal portion includes at least a portion of the inner conductor and the inner insulator and the distal portion includes a conductive member coupled to a distal end portion of the inner conductor; and

a choke disposed on the feedline and including an inner dielectric layer and an outer conductive layer, wherein the outer conductive layer is shorted to the outer conductor of the feedline and a distal end of the inner dielectric layer extends distally from a distalmost end of the outer conductive layer.

31. The microwave antenna assembly according to claim **30**, further comprising a sheath enveloping the radiating portion and defining a fluid chamber for receiving a fluid.

32. The microwave antenna assembly according to claim **31**, further comprising a tapered distal tip directly coupled to a distal end of the sheath.

33. The microwave antenna assembly according to claim **31**, further comprising a connection hub including a cable connector coupled to the feedline, an inlet fluid port through which fluid is delivered to the fluid chamber, and an outlet fluid port through which fluid is removed from the fluid chamber.

34. The microwave antenna assembly according to claim **33**, further comprising an inflow tube in fluid communication with the inlet fluid port and at least partially disposed within the fluid chamber.

35. The microwave antenna assembly according to claim **30**, wherein the gap is disposed longitudinally off-center of the unbalanced dipole antenna.

36. The microwave antenna assembly according to claim **30**, wherein an outermost diameter of the conductive member is greater than an outermost diameter of the inner conductor.

37. A microwave antenna assembly comprising:

a feedline including an inner conductor, an outer conductor, and an inner insulator disposed between the inner and outer conductors;

a radiating portion including an unbalanced dipole antenna having a proximal portion, a distal portion, and a gap disposed between the proximal and distal portions, wherein the proximal portion includes at least a portion of the inner conductor and the inner insulator and the distal portion includes a conductive member coupled to a distal end portion of the inner conductor;

a choke disposed on the feedline and including an inner dielectric layer and an outer conductive layer, wherein the outer conductive layer is shorted to the outer conductor of the feedline and the inner dielectric layer extends distally from a distalmost end of the outer conductive layer; and

a sheath enveloping the radiating portion and defining a fluid chamber for receiving a fluid.

38. The microwave antenna assembly according to claim **37**, wherein a length of the distal portion is different from a length of the proximal portion.

39. The microwave antenna assembly according to claim **37**, wherein an outermost diameter of the conductive member is greater than an outermost diameter of the inner conductor.

40. The microwave antenna assembly according to claim **37**, wherein the gap is disposed longitudinally off-center of the unbalanced dipole antenna.

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